

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5102

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Preethi R (192511352) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

R. Preethi
Signature of the student:

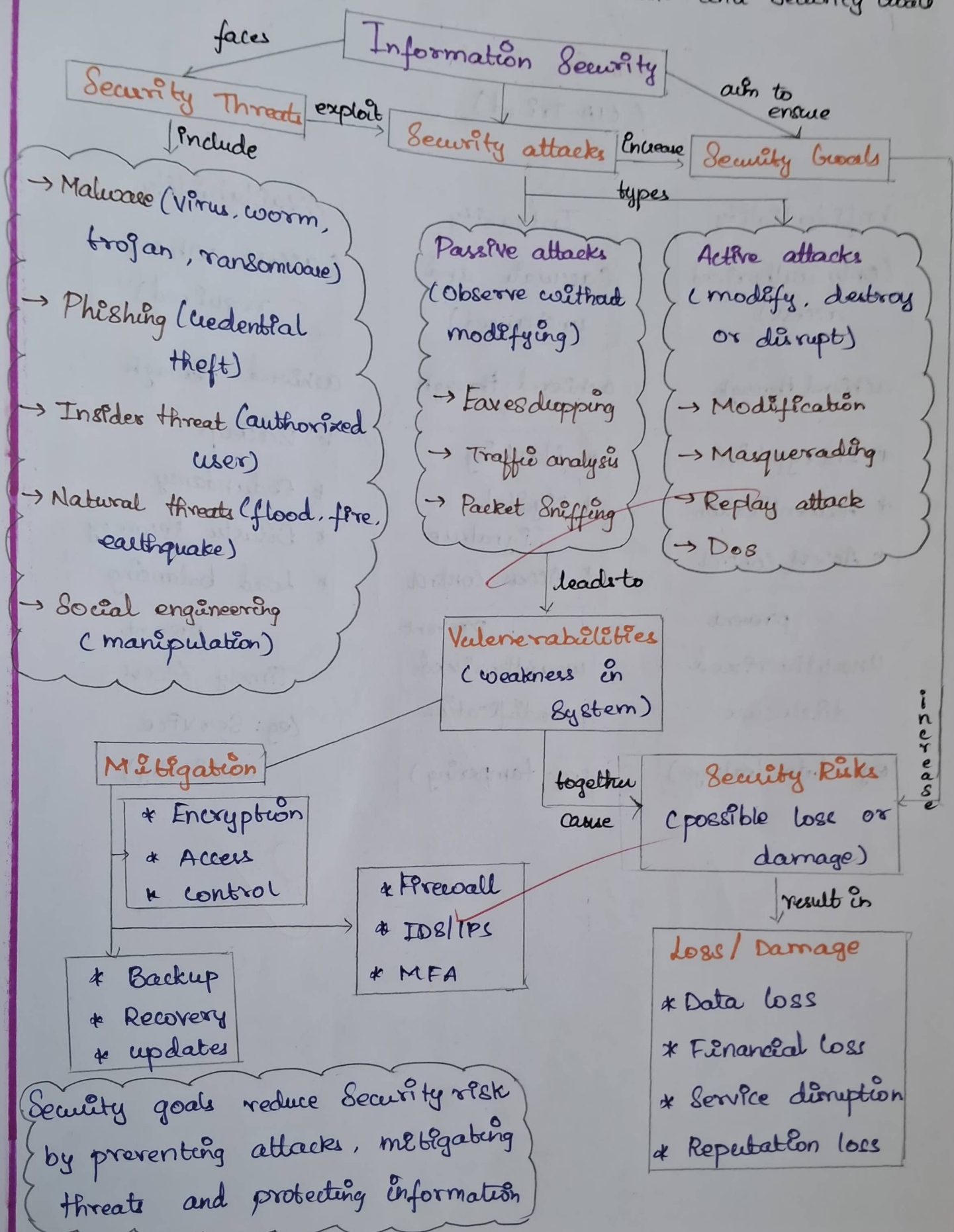
99/100

CONCEPT MAPPING

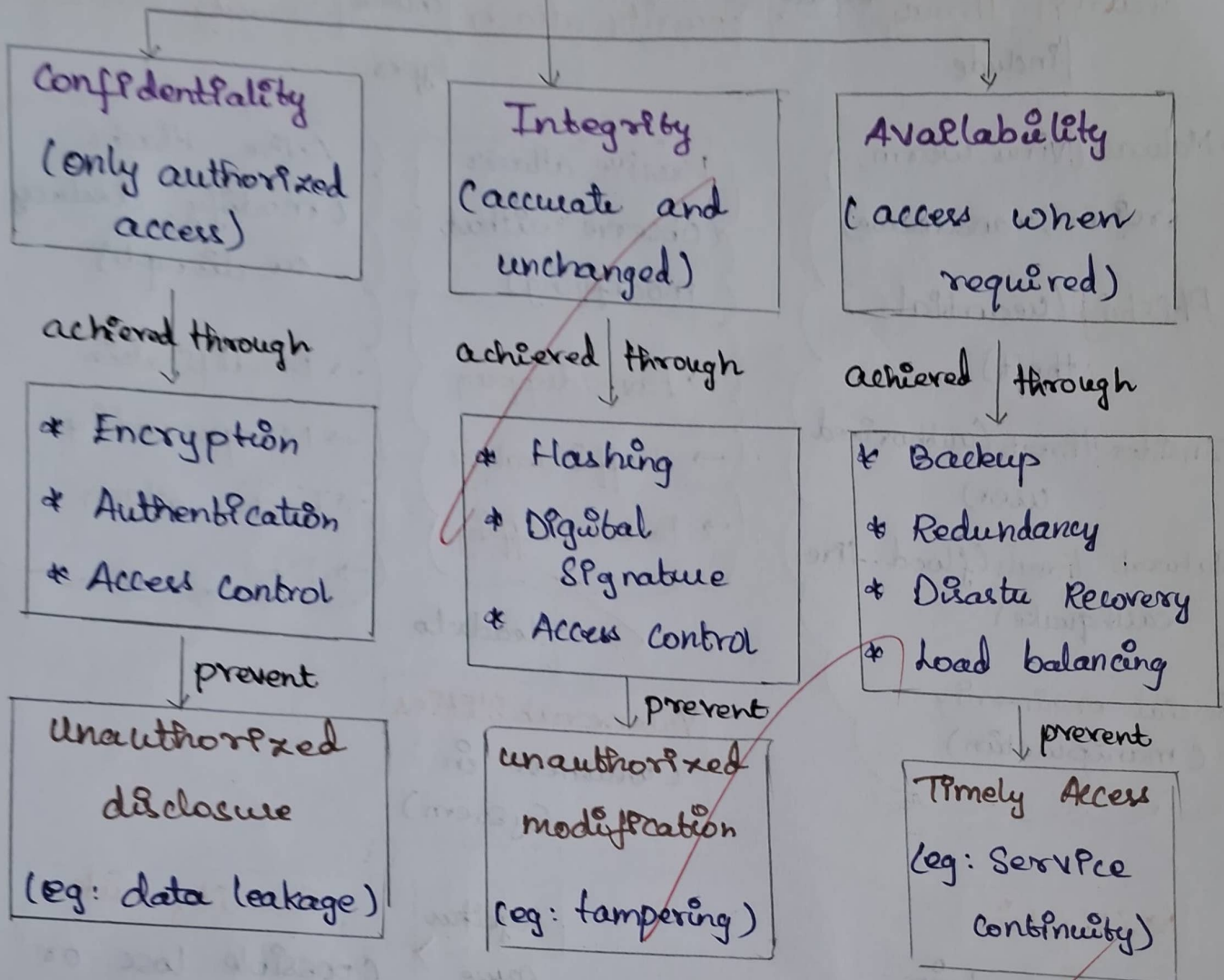
1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
 - a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why
4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each
5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Concept Map - Security attacks, Threat, Risk and Security Goals



Security Goal (CIA Triad)



2. Caesar, Monoalphabetic, Vigenere and Hill cipher

Substitution ciphers
(replace plaintext symbol with other symbols)

Caesar cipher
(Shift based)

work by

- * Shifting each letter by a fixed value
- * Single key

example

plain: ABCDE
key: 3
cipher: DEFGH

Security level

Low

Monoalphabetic cipher
(one to one mapping)

work by

- * Fixed substitution
- * Same letter → same letter each time

example

plain: ABCDE
cipher: QWERT

Security level

Low/moderate

Vigenere cipher
(multiple alphabets)

work by

- * Key word
- * Different alphabet for each position

example

plain: ATTACK
cipher: KEYKEY
cipher: KFSKWL

Security level

Moderate

Hill cipher
(matrix based)

work by

- * Matrix multiplication
- * Encrypt block of letters

example

plain: ACT
key: $\begin{bmatrix} 2 & 3 \\ 2 & 5 \end{bmatrix}$
cipher: POH

Security level

Higher complexity

Not secure by itself

- * Very simple
- * Easy to implement
- * Low computation

- * Larger key space
- * More complex than Caesar

- * Harder to find frequency patterns
- * Stronger than

- * Only 26 possible shifts
- * Brute force attack
- * Frequency analysis

- * Frequency pattern remain
- * Vulnerable to frequency analysis

- * Repeated key patterns
- * Little variation
- * frequency analysis

- * Encrypts multiple letters
- * Uses linear algebra
- * More complex
- * Matrix calculation
- * Key matrix must be invertible (mod 26)
- * Known plaintext attack

All are classical ciphers and provide security

Confidentiality, but modern system use stronger algorithm (eg: AES, ChaCha20)

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 24	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

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